Real or Generated? A Practical Detector for AI-Generated Images and Video on Social Media

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Abstract

AI-generated images and videos are now common in social media feeds, and most people cannot reliably tell them apart from real content. We propose a lightweight detection application that takes an image or short video and returns a probability that it was AI-generated, along with a confidence level. We will compare a fine-tuned CNN baseline against a CLIP-feature classifier, test how well each generalizes to generators it never saw during training, and measure how robust they are to the compression and resizing that social platforms apply. We will also analyze the ethical and societal risks of deploying such a tool, especially the cost of false accusations against real creators.

1 Problem and motivation

Diffusion models and modern video generators can produce photorealistic content in seconds. This content is increasingly posted without disclosure, where it can spread misinformation, fake product reviews, impersonate real people, or simply mislead viewers. Platform labeling is inconsistent and depends on creators self-reporting or on metadata that is easily stripped. A detector that works on the pixels themselves would give users and moderators an independent signal.

The core difficulty is *generalization*: detectors trained on one generator often fail on newer ones (6; 3). A second difficulty is that social media re-compresses and resizes uploads, which can erase the subtle artifacts detectors rely on (2). Our project targets both issues in a realistic setting.

2 Proposed solution

We will build a detection pipeline and a simple demo application.

Image detection. We will train and compare two models: (1) a ResNet-50 pretrained on ImageNet and fine-tuned for binary real/fake classification, following Wang et al. (6); and (2) a linear classifier on frozen CLIP (4) image embeddings, which has been shown to generalize better across unseen generators (3).

Video detection. We will sample frames from each video, score each frame with the image model, and aggregate scores (mean and top-k pooling) into a single video-level prediction. This keeps the system simple and lets us reuse the image model.

Demo application. A small web app where a user uploads an image or short clip and gets back a "likely real / likely AI-generated" result with a probability score, rather than a hard yes/no verdict.

3 Feasibility and evaluation

Data. For images we will use CIFAKE (1) for fast prototyping and a subset of GenImage (7), which covers multiple generators (e.g., Stable Diffusion, Midjourney, BigGAN). For video we will use a subset of FaceForensics++ (5). All datasets are public and fit on a single GPU workstation or free cloud GPU tier.

Experiments. We will report accuracy and ROC-AUC on three tests: (1) in-distribution, training and testing on the same generators; (2) cross-generator, training on some generators and testing on held-out ones; and (3) robustness, re-testing after JPEG compression, resizing, and cropping that simulate social media uploads.

Timeline. Weeks 1–2: data collection and preprocessing. Weeks 3–4: train and evaluate image models. Week 5: video extension and robustness tests. Week 6: demo app, ethics analysis, and final report.

4 Ethical, societal, and technical implications

False positives. Wrongly labeling a real photo as AI-generated can damage a creator's or journalist's credibility. We will report false positive rates explicitly and present outputs as probabilities, not verdicts. **Arms race.** Any published detector can be used to train generators that evade it, so detection alone cannot be a complete solution; it should complement provenance standards such as content credentials. **Bias.** Detectors may perform unevenly across image sources, camera quality, or skin tones in face content; we will check performance across available subgroups where the data allows. **Overreliance.** Users may treat the tool as the final word. Our interface will state its limits clearly, including that it can be wrong on generators it has not seen.

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